

REVISION 5.1 — ACID SOIL CORRECTION, MAIN FARM

Replacement pages for Greenhouses GH-01 to GH-05 and the Open-Field Habanero · Soil pH ~4.5 · Supersedes the marked pages of Revision 5

- **What changed and why.** Revision 5 was written before any block returned a soil pH reading. It puts **urea in the planting hole** and its pH rule only covers soil that is **too alkaline**. At pH 4.5 both are wrong. Phosphorus and calcium lock up, aluminium and manganese turn toxic to root tips, and urea placed near lime is lost to the air as ammonia gas within hours. **Where these pages and Revision 5 disagree, these pages win.**
- **The good news — your solarisation window is the right place for lime.** Revision 5 already has clear plastic over the beds for 3–4 weeks before transplant. Lime incorporated at the **start** of that period gets three to four weeks of warm, moist, covered reaction — which is exactly what ordinary agricultural lime needs. Any house not yet under plastic can therefore use cheap ag lime on the normal calendar, with **no 14-day emergency protocol and no clash with the Week 1–2 NPK fertigation**. Only houses already past that point need the fast route.

Pages replaced in the printed Revision 5

PAGE	SECTION	WHAT IS SUPERSEDED
3	The five gates	Gate 0 pass condition — pH is now part of the gate , not just nematodes.
5	§A Greenhouse Bell Pepper · Week 0	Whole page. Urea removed; lime and the pH gate added at the solarisation step; one-directional pH rule fixed.
13	§B Open-Field Habanero · Week H-0	Whole page. Same changes for the field.
15	§B Second flush · After flush	Urea top-dress permitted only if 3+ weeks since any lime on that block.
16	GH-4/5 Clean-Restart · Step 3	Lime goes in with the solarisation; Gate 0 now includes pH.
26	Diagnosis Kit	New card added — acid soil / aluminium toxicity.
30	EC & pH targets / Nutrient deficiency	Both tables were one-directional. Corrected to cover acid soil.

Block status register — fill this in before ordering anything

- ▲ **Do not assume all five houses read the same.** Test each house separately, three points each, meter calibrated that morning against pH 4.0 and 7.0 buffer, 1:1 soil to distilled water, stirred, left 10 minutes. Houses under different management histories drift apart. **Over-liming is far harder to reverse than under-liming** — a drifting meter reading 4.5 on soil that is really 5.4 would cost you manganese, zinc and boron for the whole cycle.

BLOCK	PH — 3 READINGS	TEXTURE	BED AREA (M²)	CURRENT STAGE	ROUTE (A OR B)	LIME KG REQUIRED
GH-01						
GH-02						
GH-03						
GH-04						
GH-05						
OF-01 habanero						
OF-03						

- › **Lime the beds, not the paths.** In the greenhouses, measure and treat **bed area only**. On a typical house the beds are roughly half the floor area — liming the walkways doubles your cost for no agronomic return. In the open field, treat the full cropped area.

WHICH ROUTE DOES EACH BLOCK TAKE?

GATE 0 · pH

ROUTE A — NORMAL (preferred). The block is **not yet under solarisation plastic**, or you are willing to delay laying it by a day. Lime goes in with the compost at the start of the solarisation window and reacts under the plastic. **Use agricultural lime — cheap, safe, no timing conflict.**

ROUTE B — FAST (only if unavoidable). The block is **already solarised**, plastic is coming off, and transplant is under 3 weeks away. **Use hydrated lime — Ca(OH)_2 , sold as slaked or builder's lime.** It acts inside two weeks because it is already in the reactive form. It also carries real handling risk and a nitrogen conflict.

Route A — agricultural lime (dolomitic)

TEXTURE	RATE	PER 100 M² BED
Loamy sand	100–135 g/m²	10–13.5 kg
Sandy loam	165–235 g/m²	16.5–23.5 kg
Loam / clay loam	265–335 g/m²	26.5–33.5 kg

These are two-thirds of the full requirement. Apply this now, re-test at the end of solarisation, apply the balance after harvest. Finest grade available — coarse lime is nearly inert.

■ **Route B has a knock-on the team must know about.** The three-week nitrogen blackout around hydrated lime runs into the **Week 1 Day 6 NPK 20-10-10 drench and the Week 2 Day 6 NPK 15-15-15** — both ammonium-based. On a Route B block, replace those two fertigations with **Calcium Nitrate 2–3 g/L** and resume the normal NPK programme from Week 3. Route A blocks have no conflict: by transplant the lime is already 4+ weeks old.

■ **PPE for hydrated lime is not optional.** Goggles, gloves, dust mask, long sleeves. It burns eyes and wet skin. Never apply on a windy morning. Brief every worker before the bag is opened, and record the briefing in the logbook.

What will NOT fix this — do not spend money on these

Gypsum supplies calcium but has **no effect on pH at all** — it is not lime, and it is the most common wasted purchase on acid soil. Also useless on this timeline: **charcoal** (the liming power is in ash, not char — you would need 6 to 18 tonnes to match 200 kg of lime), compost alone, and liquid “soil conditioner” pH correctors, which are mostly humic acid blends with negligible neutralising value. **Wood ash** is a legitimate fallback if lime is unobtainable — hardwood only, no treated timber or rubbish burn, capped at 150 g/m², counted as roughly one-third the strength of hydrated lime.

➤ **Keep the neem cake separate from the lime.** Revision 5 puts neem cake 250 g/hole at 2 weeks before transplant. Neem cake releases its nitrogen as ammonium, so on a Route B block it sits inside the blackout window. Keep the two applications at least 10 days apart — lime first, neem cake second. On Route A this resolves itself, since the lime went in at the start of solarisation.

REPLACES PAGE 5 — \$A GREENHOUSE BELL PEPPER · WEEK 0 · Day 1–7

Gates 0 + 1

▲ UV-net side walls = passive ventilation (no operable roof vents). Airflow lowers humidity 5–10% vs a sealed house — a real disease advantage — but whitefly, thrips and aphids enter as freely as outdoors. Yellow sticky traps and the Day 2 scouting round are not optional.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
3–4 WKS BEFORE	LIME + SOLARISATION + NEEM CAKE	- Three-point pH test first. Calibrated meter, 1:1 soil to distilled water. Log all three readings per house with a photo of the meter. - NEW — Route A lime: broadcast dolomitic ag lime at the rate for your texture over bed area only. Incorporate to 15–20 cm. Irrigate. Dry lime does not react. - Cover all beds with CLEAR (transparent) polythene, pulled tight, edges sealed. Leave 3–4 weeks — PH sun drives soil to 55–70°C, killing nematodes, fungal spores, cutworms and weed seed. The same warmth and moisture drives the lime reaction. - Never black plastic (it insulates, does not heat). No air pockets. - At 2 weeks before transplant: 250 g neem cake per planting hole, mixed in. Keep it 10+ days clear of the lime. - Never Furadan (carbofuran) — banned, destroys soil biology. - End of period: lift plastic, air beds 2 days, then transplant week.	Clear plastic only, laid flat, no air gaps. Lime under plastic must go in before the sheet, never after.	GO pH logged; lime in and incorporated; clear plastic 3–4 wks; neem cake at Wk-2. HOLD Black plastic, under 3 wks, or plastic laid before liming.
Day 1	SOIL FINAL PREP + pH GATE + EC	- NEW — re-test pH, three points. This is the Gate 0 pH check. Result 5.5–7.0 → cleared. Below 5.2 → apply half the original rate again, add 10 days, do not stack a full second dose. Urea 50 g/hole (3 top) — DELETED. No urea, no ammonium NPK in the hole. Nitrogen arrives after establishment as Calcium Nitrate (see the nutrition table overleaf). - Compost 20–30 kg/bed into top 10 cm (yearly target 5 t/ha). - EC/pH every week from now. Two-way rule: irrigation water pH 5.8–6.5; if >7.0 add citric acid 5 g/10 L; if soil pH <5.5 the block is HELD. Plain-water EC 1.8–2.8 mS/cm (>3.5 = flush 60 min). - Run drip 60 min to settle beds and verify every emitter; mark and fix blocked ones.	Water not draining within 10 min = blockage. Never transplant into flooded beds — or into an unrecorded pH.	GO pH ≥5.5 logged with photo; draining well; emitters flowing. HOLD pH below 5.5, no re-test on file, waterlogged, or >5 blocked emitters.
Day 3 · 4–6 PM	PRE-PLANT FUNGICIDE	- Mancozeb 80WP 2.5 g/L (40 g / 16 L). Spray beds, floor, posts and the inside of the net walls. 4–6 PM only. Mask + gloves. - Log 'Mancozeb, FRAC M3' — this opens your FRAC rotation.	Wind-driven rain through the nets dilutes it; beds should look faintly dusty 2 h later.	GO Beds dry; 4–6 PM; calm air. HOLD Rain-laden wind through nets — wait.
Day 5 · 4–7 PM	PRE-PLANT INSECTICIDE	- Cypermethrin 10EC 1 mL/L (16 ml / 16 L). Spray all interior surfaces — net walls both faces, floor, posts, under benches. - Do this within 48 h of transplant. Doors shut overnight. Rinse knapsack 3×. - Log 'Cypermethrin, IRAC 3A' — the next insecticide MUST be a different group.	Net walls let insects recolonise fast — don't spray a week early and leave doors open.	GO 4–7 PM; doors shut; logged 3A. HOLD Strong wind/rain — reschedule.
Day 6 · 4–6 PM	TRANSPLANT + VENTILATION PROTOCOL	- Route B blocks: confirm 14 days have passed since the hydrated lime. Route A blocks: no wait, the lime is already 4+ weeks old. - Transplant 4–6 PM only. 1 seedling/hole at 30 cm, firm gently, water each row as you plant. - Ventilation (daily, whole cycle): open all doors 10 AM–2 PM to flush the mid-day humidity spike the passive net walls can't clear alone; close at 2 PM to hold evening warmth. - Install yellow sticky traps: 1 per 6 m² at plant height. - Drip 30 min after planting. Logbook: today = WEEK 0, DAY 1 — every future week counts from here.	Seedlings must be hardened 7+ days; soft lush ones collapse. Fresh hydrated lime burns new roots.	GO Overcast PM; hardened; soil pre-moist; traps up; lime window cleared. HOLD Mid-day heat; unhardened; dry soil; under 14 d since hydrated lime.
Day 7	WATER + DAY-1 CHECK	- Drip 6:30 AM 20 min · 1 PM 15 min (skip only if net walls show condensation AND soil moist 5 cm deep) · 5 PM 20 min. - Walk every house. Day 1–3 mid-day wilt with evening recovery = normal transplant shock. - Still collapsed by 6 PM from Day 3 on: pull one and read the roots — white firm = water only; brown slimy = rot; red/brown galls = nematode; stubby, thickened, dead-tipped but NO galls = acid soil (see the new diagnosis card).	A persistent wilter's roots tell you the cause. Read them, don't guess.	GO Mild mid-day wilt, evening recovery. HOLD Full wilt unrecovered by 6 PM Day 3 — investigate roots now.

REPLACES PAGE 13 — \$B OPEN-FIELD HABANERO · WEEK H-0 · Land Prep & Transplant

Gates 0 + 1

▲ **Open field = no overhead cover.** Rain washes off spray (reapply within 4 h of >15 mm rain); full sun dries soil faster; and bare soil splashes Phytophthora onto crowns — **mulch to 12 cm** is essential at 30 cm spacing. Wind helps airflow, so disease pressure is a little lower than the greenhouse. **Rain also leaches lime faster in the open field than under greenhouse cover — expect to re-lime the field a season sooner than the houses.**

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
3–4 WKS BEFORE	LIME + SOLARISATION + NEEM CAKE + COMPOST	• Three-point pH test first. Log the readings with a photo. • NEW — lime the full cropped area at the Route A rate for your texture. Incorporate to 15–20 cm along with the compost. Irrigate. • Clear polythene over the whole field 3–4 weeks (open-field sun reaches 65–70°C — one cycle drops nematodes 70–80% in the top 20 cm). The lime reacts under it. • Neem cake 250 g/hole at 2 weeks before, 10+ days clear of the lime; compost 5 t/ha (2 kg/m²) incorporated before solarisation. No Furadan. • After: lift plastic, air 2 days, land prep. Irrigation water pH 5.8–6.5, EC 1.8–2.8.	Transparent plastic only; solarisation is even stronger in open sun.	GO pH logged; lime in; clear plastic 3–4 wks; neem + compost in. HOLD Black plastic, <3 wks, or no pH on file.
Day 1–3	LAND PREP + pH GATE + 12 cm MULCH	• NEW — re-test pH, three points. Gate 0 pH check. 5.5–7.0 → cleared. Below 5.2 → half rate again, add 10 days. • Urea 50 g/hole mixed in — DELETED. Nitrogen arrives as Calcium Nitrate through the field tank after establishment. • Mulch (dry grass / rice straw / coco peat) to 12 cm across the ENTIRE field incl. between plants — at 30 cm spacing there is almost no bare soil, cover it all. • Run drip 60 min to settle; verify emitters.	Mulch under 8 cm in heavy rain = Phytophthora crown-rot outbreak.	GO pH ≥5.5 logged; mulch 12 cm; drip flowing; neem applied. HOLD pH below 5.5; mulch below 8 cm — top up.
Day 5 · 4–6 PM	TRANSPLANT HABANERO	• Route B blocks: confirm 14 days since hydrated lime before a seedling goes in. • Transplant 4–6 PM only. 30 × 30 cm, each plant over an emitter, straight rows. Water each row as you plant. • Install 100 cm stakes now (habanero gets bushy and fruit-heavy). • Logbook: transplant date = HABANERO WEEK H-0, Day 1.	Habanero resists shock better than bell pepper but still wilts in mid-day sun.	GO 4–6 PM; hardened; stakes ready; lime window cleared. HOLD Mid-day transplant under full sun; <14 d since hydrated lime.
Day 6–7	WATERING + DAY-1 CARE	• Field irrigation, all weeks: 6:30 AM 30 min (dry) / 20 min (rainy); 5 PM 20 min (skip if >15 mm rain today). After >25 mm rain, skip next morning entirely. • Walk the field, flag any plant not upright. Day 1–3 wilt normal; not recovered by Day 3 evening → read roots (galls = nematode; stubby and dead-tipped without galls = acid soil).	Open-field soil dries fast in PH's ~33°C sun; the 6:30 AM run builds a buffer before peak heat.	GO Drip twice daily; plants recovering. HOLD Full drip after >25 mm rain — root suffocation.

› **Also amend page 15 (Second flush · After flush).** Revision 5 says *urea top-dress 20 g/plant around the drip zone*. That stays — but **only if 3+ weeks have passed since any lime was applied to that block**. Inside three weeks, substitute Calcium Nitrate through the field tank at the same nitrogen equivalent, and resume urea on the following flush.

AMENDED GATE 0, REFERENCE TABLES & NEW DIAGNOSIS CARD

RC1 · RC4

● **Gate 0 amendment — the structural change that matters most (replaces the Gate 0 row on page 3).** Revision 5's Gate 0 pass condition is a soil and nematode report only. As written, the team can clear Gate 0 with a pH of 4.5 printed on that report and plant the block legally. **Gate 0 now reads: soil + nematode report returned clear AND soil pH between 5.5 and 7.0, three-point test on file, signed off by the agronomist. No seedling goes in until both are green.** If pH is not a hard stop, it is not protected — which is exactly the lesson RC1 taught about nematodes.

Corrected EC & pH targets (replaces page 30)

MEASURE	TARGET
Soil pH — Gate 0	5.5–7.0. Below 5.5 = block HELD, lime and re-test.
pH irrigation water	5.8–6.5 (if >7.0 add citric acid 5 g/10 L)
EC plain water	0.3–0.8 mS/cm
EC fertigation (veg)	1.8–2.8 mS/cm
EC fertigation (fruit)	2.0–3.0 mS/cm
EC >3.5 mS/cm	Stop — flush 60 min before fertigating

Nutrient deficiency — corrected rows (page 30)

SYMPTOM	CAUSE	ACTION
Interveinal yellow, middle leaves	Iron / Manganese	Check pH both ways — above 7.0 locks Fe/Mn, acidify. Below 5.5 they turn toxic instead — lime.
Purpling on older leaves; stunted, no galls	Phosphorus locked by acid soil	Test pH. Place starter P near the root zone; broadcast P is wasted below 5.5.
Blossom-end rot despite a full Ca programme	Ca unavailable at low pH	Test pH. Ca Nitrate 1.8 kg/1,000 L holds the crop; lime fixes the cause next cycle.

Replacement nutrition — what goes in instead of the urea

WHEN	PRODUCT	RATE	WHY THIS ONE
Day 10–14 after transplant	Calcium Nitrate — fertigation	2–3 g/L weekly (GH 1,000 L tank / field 500 L tank)	Nitrate nitrogen is immediately available, does not burn young roots, and does not fight the lime. Below pH 5.5 nitrification is sluggish anyway, so ammonium nitrogen would sit unused while acidifying further. The calcium is the second reason: Ca availability is poor in acid soil and blossom-end rot is otherwise close to guaranteed.
At planting	Starter P, placed near the root zone	Per label	Phosphorus is heavily fixed below pH 5.5. Broadcast P on these blocks is largely wasted money; placed P reaches the roots.
Route B blocks only	Substitute for Wk 1 & Wk 2 NPK	Ca Nitrate 2–3 g/L	NPK 20-10-10 and 15-15-15 are ammonium-based and fall inside the 3-week blackout. Resume the normal NPK programme from Week 3.

SOIL NEW CARD — Acid soil / aluminium & manganese toxicity (pH below 5.5) · add to page 26

CAUSE

Below pH 5.5 aluminium and manganese become soluble and toxic to root tips, while phosphorus locks into unavailable compounds and calcium and magnesium leach away. The plant starves in soil that may contain adequate nutrients. Endemic to the high-rainfall Niger Delta, where decades of leaching strip base cations from the topsoil.

DETECTION

Stunted, pale, nutrient-starved plants with **stubby, thickened, poorly branched roots and dead root tips — but no galls**. That absence is the distinguishing test against nematode. Purpling on older leaves indicates the phosphorus lockup. Confirm with a three-point pH test, calibrated meter, 1:1 soil to water.

PREVENTION

Gate 0 now tests pH as well as nematodes. Lime during the solarisation window, when ag lime has time to work properly, rather than reaching for an emergency correction a week before transplant. Re-test every cycle — acid soil returns, faster under ammonium fertiliser and heavy rain, and faster in the open field than under greenhouse cover.

TREATMENT

Mid-cycle there is **no fast fix** — the roots are already damaged and lime cannot be safely incorporated around a standing crop. Hold the crop with nitrate-based nutrition and calcium nitrate fertigation, place phosphorus rather than broadcasting it, and correct the block at the post-harvest window. Like galled roots, this is a Gate 0 problem: it is prevented, not cured.

▲ **Why this card matters more than it looks.** Acid-soil stunting presents as **stunted, mid-day wilting, nutrient-starved plants that no spray fixes** — word for word the root-knot nematode row on page 26. A team trained to pull a plant and look for galls will find none, and then start guessing. **That is RC4 with a new cause.** The root check on Day 7 of Week 0 now names both possibilities so nobody has to guess between them.